

Changyu (Thomason) Zhao

608-886-7470 | czhao249@wisc.edu | [linkedin.com/in/changyu-zhao](https://www.linkedin.com/in/changyu-zhao) | github.com/ThomasonZhao

EDUCATION

University of Wisconsin - Madison

Bachelor of Science in Computer Science and Math

Madison, Wisconsin

Sep. 2021 – May 2025

EXPERIENCE

Tencent Spark Early Career Camp

Cybersecurity Researcher Intern

June 2021 – Present

Shenzhen, China

- Through source code review to find the deserialization and sandbox escape vulnerability of modified Vijos Online Judge (OJ) system. Successfully attacked by gaining the reverse root privileged shell under docker container.
- Escalated privilege to root by escaping from privileged docker container by abusing CVE-2019-5736.
- Utilized IDA to reverse engineering the router firmware and locate buffer overflow mentioned in CVE-2018-8879. Applied Returned Oriented Programming (ROP) skills to get reverse shell and fully control the router.

PROJECTS

Tenda AC18 AC1900 Router Exploitation | IDA, GDB, QEMU, Python

May 2022 – Aug.2022

- Exploited unfixed Common Vulnerabilities and Exposures (CVE) on Tenda AC18 router firmware.
- Applied qemu-system to emulate arm32 router firmware on x86-64 machine for systematic debugging.
- Developed decompiled code level automated stack overflow and command injection detection plug-in for IDA.
- Achieved Remote Code Execution (RCE) through command injection and buffer overflow vulnerability.
- Discovered unfixed CVEs: CVE-2018-16334, CVE-2018-5767, CVE-2018-5768, CVE-2018-5770, CVE-2020-15916, CVE-2022-28556, CVE-2022-28557.

Windows Calculator Reverse Engineering | IDA, Windbg, Frida

Jan. 2022 – April 2022

- Exploited Win10 & Win32 Calculator to return absolutely wrong arithmetic result, for example: $1 + 1 = 3$.
- Utilized IDA and Windbg to reverse engineering and do both static and dynamic analysis on the calculator binary.
- Utilized Frida to apply hooks on the calculator's library functions to hijack arbitrary arithmetic operation.

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team

Team member

Sep. 2021 – Present

China

- DEF CON CTF Finals, Rank 12/1282 world wide.
- Competed in more than 20 48-hour CTFs since 2021; Specialized in Binary Exploitation & Reverse Engineering

pwn.college

Independent Learning

Dec. 2021 – Jul. 2022

Temp, Arizona

- Earned yellow belt and blue belt (Finish all 472 challenges in the dojo instance). Current ranking 43/2500+.
- Learned binary exploitation skills: program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race condition, and kernel security.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, x86-Assembly JavaScript, HTML/CSS, $\text{\LaTeX}$

Binary Security Tools/Framework: IDA/Ghidra, GDB, Windbg, Angr, Frida, Elastic Search, Joern

Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux